INCH-POUND

MS16632C
1 May 2012
SUPERSEDING
MS16632B
30 April 1990

DETAIL SPECIFICATION SHEET

RING, RETAINING, EXTERNAL, CRESCENT
(REDUCED SECTION TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-R-21248.

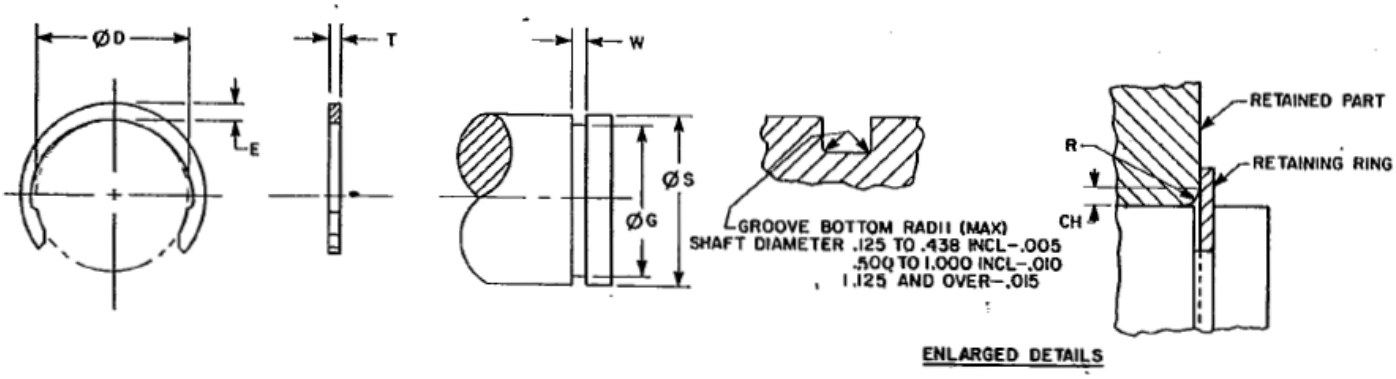


FIGURE 1. RING, RETAINING.

AMSC N/A

FSC 5325

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

TABLE I. DIMENSIONS.

ØS SHAFT (REF)		ØD FREE		E LARGE HEIGHT		T 1/ THICKNESS		ØG RECOMMENDED GROOVE (REF)		W WIDTH		R 3/ OF RETAINED PART (REF)	CH 3/
INCH	MM	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MAX	MAX
.125	3.2	.102	+.002	.031		.015		.106	±.0015	.020	+.002	.014	.011
.156	4.0	.131	-.004	.037	±.003	.015		.135	.0015	.020	-.000	.018	.014
.188	4.8	.161		.042		.015		.165	FIM 2/	.020		.021	.016
.219	5.6	.187		.044		.025		.193		.030		.021	.016
.236	6.0	.203		.046		.025		.208		.030		.022	.017
.250	6.4	.211		.050		.025		.220		.030		,023	.018
.281	7.1	.242		.052		.025		.247	±.002	.030		.021	.016
.312	7.9	.270	+.003	.053	±.004	.025		.276	.002	.030		.024	.018
.375	9.5	.328	-.005	.060		.025		.335	FIM 2/	.030		.026	.020
.406	10.3	.359		.063		.025		.364		.030		.027	.021
.438	11.1	.386		.065		.025	±.002	.393		.030		.029	.022
.500	12.7	.441		.070		.035		.450		.040	+.003	.030	.023
.562	14.3	.497	±.006	.078		.035		.507		.040	-.000	.033	.025
.625	15.9	.553		.081		.035		.563		.040		.033	.025
.688	17.5	.609		.086		.042		.619		.047		.034	.026
					±.005				±.003				
.750	19.0	.665		.090		.042		.676	.004	.047		.036	.027
.812	20.6	.721	±.007	.097		.042		.732	FIM 2/	.047		.038	.029
.875	22.2	.777		.105		.042		.789		.047		.040	.031
.938	23.8	.830		.112		.042		.843		.047		.043	.033
1.000	25.4	.887		.120		.042		.900		.047		.046	.035
1.125	28.6	.997		.135		.050		1.013		.056		.052	.040
1.250	31.7	1.110	±.008	.150		.050		1.126	±.004	.056	+.004	.057	.044
1.375	34.9	1.220		.165	±.007	.050		1.237	.005	.056	-.000	.062	.048
1.500	38.1	1.331		.180		.050		1.350	FIM 2/	.056		.069	.053
1.750	44.4	1.555	±.010	.210		.062	±.003	1.576	±.005	.068		.081	.062
2.000	50.8	1.777		.240		.062		1.800	.005 FIM 2/	.068		.091	.070

1/ T - THICKNESS "T" APPLIES TO UNPLATED RINGS. FOR CORROSION RESISTANT STEEL AND PLATED RINGS, +.002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i.e. ±.002 SHOULD BE +.004/-.002.

2/ FIM - (FULL INDICATOR MOVEMENT) IS THE MAXIMUM ALLOWABLE DEVIATION OF CONCENTRICITY BETWEEN THE GROOVE AND THE SHAFT.

3/ R AND CH - RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RINGS. ALLOWABLE THRUST LOADS FOR RINGS RETAINING PARTS WITH MAXIMUM CORNER RADII OR CHAMFERS ARE LSTED IN TABLE VI ON PAGE 8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. CLASSIFICATION: RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE I, CLASS 6 OF THE PROCUREMENT SPECIFICATION.

2. MATERIAL:

(a) CARBON SPRING STEEL, GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A 568/A568M OR ASTM A684/A684M.
(b) CORROSION RESISTANT STEEL, IN ACCORDANCE WITH SAE AMS 5520 (UNS S15700).
(c) BERYLLIUM COPPER, ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH ASTM B194.

3. HARDNESS

TABLE II. HARDNESS.

ØSHAFT (REF)	CARBON STEEL	CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.125 TO .438 INCL	84.5-87.0 HRI5N 1/	82.5-86 HRI5N 1/	77-82 HRI5N 1/
.500 TO .625 INCL	66.5-71.5 HR30N 1/	63-69.5 HR30N 1/	54-62 HR30N 1/
.688 TO 1.000 INCL	52.5-58.5 HR45N 1/	48-56 HR45N 1/	36-47 HR45N 1/
1.125 TO 2.000 INCL	48-53 HRC	44-51 HRC	34-43 HRC

1/ USE HIGHEST SCALE TO PROVIDE SECTION WIDTH EQUAL TO OR GREATER THAN 5 TIMES THE BRAILE IMPRESSION DIAMETER.

4. PROTECTIVE FINISH OR TREATMENT:

a. CARBON STEEL - SHALL BE AS SPECIFIED (SEE TABLE Ill OR IV)

1. CADMIUM PLATE IN ACCORDANCE WITH SAE AMS-QQ-P-416, TYPE II, CLASS 3 OR ASTM B 696, TYPE II, CLASS 5.
2. ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn 5 or ASTM B 695, TYPE II, CLASS 5.
3. PHOSPHATE COAT IN ACCORDANCE WITH MIL-DTL-16232, TYPE Z, CLASS 2.

b. CORROSION RESISTANT STEEL - SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH SAE AMS2700.

5. PART NUMBER: THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE Ill OR IV.

EXAMPLE: MS16632-1100 IS THE PART NUMBER FOR A CARBON STEEL, CADMIUM PLATE, EXTERNAL, CRESCENT RETAINING RING FOR USE ON A 1.000 SHAFT DIAMETER.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

TABLE III. DASH NUMBERS FOR MS16632.

Ø S SHAFT (REF)	1/ CARBON STEEL CADMIUM PLATE DASH NO	1/ CARBON STEEL ZINC COAT DASH NO	1/ CARBON STEEL PHOSPHATE COAT DASH NO	STEEL CORROSION RESISTANT DASH NO	1/ BERYLLIUM COPPER DASH NO
.125	-1012	-2012	-3012	-4012	-5012
.156	-1015	-2015	-3015	-4015	-5015
.188	-1018	-2018	-3018	-4018	-5018
.219	-1021	-2021	-3021	-4021	-5021
.236	-1023	-2023	-3023	-4023	-5023
.250	-1025	-2025	-3025	-4025	-5025
.281	-1028	-2028	-3028	-4028	-5028
.312	-1031	-2031	-3031	-4031	-5031
.375	-1037	-2037	-3037	-4037	-5037
.406	-1040	-2040	-3040	-4040	-5040
.438	-1043	-2043	-3043	-4043	-5043
.500	-1050	-2050	-3050	-4050	-5050
.562	-1056	-2056	-3056	-4056	-5056
.625	-1062	-2062	-3062	-4062	-5062
.688	-1068	-2068	-3068	-4068	-5068
.750	-1075	-2075	-3075	-4075	-5075
.812	-1081	-2081	-3081	-4081	-5081
.875	-1087	-2087	-3087	-4087	-5087
.938	-1093	-2093	-3093	-4093	-5093
1.000	-1100	-2100	-3100	-4100	-5100
1.125	-1112	-2112	-3112	-4112	-5112
1.250	-1125	-2125	-3125	-4125	-5125
1.375	-1137	-2137	-3137	-4137	-5137
1.500	-1150	-2150	-3150	-4150	-5150
1.750	-1175	-2175	-3175	-4175	-5175
2.000	-1200	-2200	-3200	-4200	-5200

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

TABLE IV. SUBSTITUTION TABLE, CROSS REFERENCE PART NUMBERS

Ø S SHAFT (REF)	INACTIVE	SUBSTITUTE 1/	SUBSTITUTE 1/	SUBSTITUTE 1/
	CARBON STEEL	CARBON STEEL CADMIUM PLATE	CARBON STEEL ZINC COAT	CARBON STEEL PHOSPHATE COAT
	MS16632	MS16632	MS16632	MS16632
.125	-12	-1012	-2012	-3012
.156	-15	-1015	-2015	-3015
.188	-18	-1018	-2018	-3018
.219	-21	-1021	-2021	-3021
.236	-23	-1023	-2023	-3023
.250	-25	-1025	-2025	-3025
.281	-28	-1028	-2028	-3028
.312	-31	-1031	-2031	-3031
.375	-37	-1037	-2037	-3037
.406	-40	-1040	-2040	-3040
.438	-43	-1043	-2043	-3043
.500	-50	-1050	-2050	-3050
.562	-56	-1056	-2056	-3056
.625	-62	-1062	-2062	-3062
.688	-68	-1068	-2068	-3068
.750	-75	-1075	-2075	-3075
.812	-81	-1081	-2081	-3081
.875	-87	-1087	-2087	-3087
.938	-93	-1093	-2093	-3093
1.000	-100	-1100	-2100	-3100
1.125	-112	-1112	-2112	-3112
1.250	-125	-1125	-2125	-3125
1.375	-137	-1137	-2137	-3137
1.500	-150	-1150	-2150	-3150
1.750	-175	-1175	-2175	-3175
2.000	-200	-1200	-2200	-3200

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. RECOMMENDED DESIGN LIMITATIONS AND USAGE.

(a) INTENDED USE- TO PROVIDE SHOULDERS FOR POSITIONING AND MAINTAINING MACHINE PARTS ON SHAFTS WHICH ARE AXIALLY INACCESSIBLE IN ASSEMBLY THEY ARE APPLIED RADIALLY AND BECAUSE OF DEEP GROOVES, HAVE HIGH THRUST CAPACITY. THEY ARE OF ADVANTAGE WHERE FAST ASSEMBLY FOR MASS PRODUCTION LINES IS ESSENTIAL AND WHERE COMPARATIVELY SMALL CLEARANCE DIAMETERS ARE DESIRABLE. THE USE OF THE FOLLOWING FORMULAS ARE BASED ON THE FACT THAT THE RING WILL NOT FAIL IN COMPRESSION.

LIMITATION ON USE- THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON, ETC.

WARNING - RINGS SHOULD NOT BE OVER EXPANDED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE. IF RING HAS PLAY BETWEEN THE GROOVE DIAMETER AND THE INSIDE RING DIAMETER THIS INDICATES THAT THE RING HAS BEEN OVER EXPANDED, (PROVIDING GROOVE HAS BEEN MACHINED TO RECOMMENDED DIMENSIONS).

FOR APPROXIMATE SAFETY RPM LIMITS SEE TABLE V.

TABLE V. APPROXIMATE SAFETY RPM LIMITS.

Ø SHAFT(INCHES)		.250	.500	1.000	2.000
CARBON STEEL AND CORROSION RESISTANT STEEL	RPM LIMIT	60,000	25,000	12,500	6,000
BERYLLIUM COPPER	RPM LIMIT	38,000	16,000		

(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RINGS. ABUTTING COMPONENTS TO HAVE SHARP CORNERS=

$$P = \frac{\pi STX}{2F}$$

WHERE:
P = ALLOWABLE THRUST LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
T = RING THICKNESS (INCHES)
X = ULTIMATE SHEAR STRENGTH OF THE RING MATERIAL (PSI) 1/
F = FACTOR OF SAFETY

A SAFETY FACTOR, F= 4, IS RECOMMENDED SINCE THE RING UNDER LOAD IS SUBJECTED NOT ONLY TO PURE SHEAR STRESSES, BUT ALSO TO BENDING STRESSES.

(c) ALLOWABLE LOAD CAPACITY OF GROOVE WALL

$$P = \frac{\pi GdY}{2F}$$

WHERE:
P = ALLOWABLE COMPRESSION LOAD (POUNDS)
G = GROOVE DIAMETER (INCHES)
d = GROOVE DEPTH (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
F = FACTOR OF SAFETY

TO INSURE A SAFE WORKING LOAD, A SAFETY FACTOR, F= 2, IS RECOMMENDED.

1/ X = 120,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS UP TO AND INCLUDING .625 INCH SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .688 INCH AND OVER SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 110,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF ALL SHAFT DIAMETERS AND OF BERYLLIUM COPPER.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

(d) MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT =

$Z = 2d$

WHERE: Z = MINIMUM DISTANCE BETWEEN OUTER GROOVE AND END OF SHAFT (INCHES)
d = GROOVE DEPTH (INCHES)

(e) DIFFERENTIAL ROTATION =

THE CONDITIONS UNDER WHICH A RETAINING RING MAY BE USED WHEN ADJACENT PARTS ROTATE RELATIVE TO IT, FALL INTO TWO CATEGORIES:

1. WHERE NO THRUST IS EXERTED BY ADJACENT PART; IN THIS CASE, DIFFERENTIAL ROTATION OF RING AND ADJACENT PART CREATES NO ELEMENT OF RISK IN THE APPLICATION OF THE RINGS BECAUSE NO FUNCTIONAL TORQUE IS EXERTED BY THE MACHINE PART ON THE RING.

2. CONSIDERATION MUST BE GIVEN TO THE MAGNITUDE OF THE THRUST INVOLVED. THE FRICTION MOMENT MAY NOT EXCEED THE BENDING MOMENT, WHICH THE RING CAN TOLERATE WITHOUT RELEASING ITS PRESSURE AGAINST THE BOTIOM OF THE GROOVE, FORMULATED AS FOLLOWS:

$fPN \leq \frac{sTE^2}{18}$ OR $P \leq \frac{sTE^2}{f18N}$

WHERE: P = ALLOWABLE THRUST LOAD EXERTED BY ADJACENT PART (POUNDS)
f = COEFFICIENT OF FRICTION
s = WORKING STRESS OF RING UNDER MAXIMUM EXPANSION (PSI) 2/
T = RING THICKNESS (INCHES)
E = LARGEST WIDTH SECTION OF RING (INCHES)
N = NEUTRAL RING DIAMETER (INCHES) = FREE DIAMETER PLUS 1 1/2 E DIMENSION

IN SUCH CASES WHERE DIFFERENTIAL ROTATION OCCURS, THE CALCULATION SHOULD BE BASED ON THE MAXIMUM POSSIBLE VALUE OF THE COEFFICIENT OF FRICTION.

(f) IMPACT CAPACITY OF RING OR GROOVE WALL =

$I_R = \frac{PT}{2}$ - FOR THE RING (INCH POUNDS), ABUTTING COMPONENTS TO HAVE SHARP CORNERS.

$I_G = \frac{Pd}{2}$ - FOR THE GROOVE (INCH POUNDS)

WHERE: P = ALLOWABLE THRUST LOAD OF RING OR GROOVE (POUNDS)
T = RING THICKNESS (INCHES)
I_G = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS)
d = GROOVE DEPTH (INCHES)
I_R = IMPACT CAPACITY OF RING (INCH POUNDS)

(g) LOAD CAPACITY, WITH THE RETAINED PART RADIUSED OR CHAMFERED =

WHEN THE RADIUS OR CHAMFER OF THE RETAINED PART DOES NOT EXCEED THE MAXIMUM RADIUS ALLOWED FOR THE BOTTOM OF THE RING GROOVE, THE LESSER LOAD CAPACITY COMPUTED FROM THE FORMULAS ON PAGES 6 AND 7 WILL APPLY. THE CORNER RADII AND CHAMFERS LISTED ON TABLE I WERE CHOSEN AS LARGE AS POSSIBLE FOR THE RING SIZES INVOLVED AND ARE RELATED TO THE MAXIMUM THRUST LOADS LISTED IN TABLE VI. IF THE CORNER RADII OR CHAMFERS ARE SMALLER THAN THOSE LISTED, THEN THE THRUST LOADS INCREASE PROPORTIONALLY, IN ACCORDANCE WITH THE FOLLOWING FORMULAS:

$P^1 = \frac{P\,CH}{CH^1}$ OR $P^1 = \frac{PR}{R^1}$

WHERE: p1 = NEW ALLOWABLE THRUST LOAD
P = LISTED ALLOWABLE THRUST LOAD
CH^1 = NEW (SMALLER) CHAMFER
CH = LISTED CHAMFER
R^1 = NEW (SMALLER) CORNER RADIUS
R = LISTED CORNER RADIUS

2/ S = 250,000 PSI WORKING STRESS FOR RINGS OF CARBON STEEL OR CORROSION RESISTANT STEEL.
S = 180,000 PSI WORKING STRESS FOR RINGS OF BERYLLIUM COPPER.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.

LIMIT LOADS LISTED IN TABLE VI ARE BASED ON RINGS OF CARBON STEEL OR CORROSION STEEL (WORKING STRESS 250,000 PSI). IF THE ALLOWABLE GROOVE CAPACITY LOADS AS CALCULATED BY USING THE FORMULA ON PAGE 7 ARE LESS, THEN THEY SHOULD BE USED.

TABLE VI. LIMIT LOADS.

S NOMINAL RING SIZE		ALLOWABLE THRUST LOAD FOR RING ASSEMBLIES WITH PARTS HAVING MAXIMUM CORNER RADII OR CHAMFERS	
FROM	TO	CARBON STEEL OR CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.125		90 LB	65 LB
.156		105 LB	75 LB
.188		110 LB	80 LB
.219		260 LB	185 LB
.236		275 LB	200 LB
.250		290 LB	210 LB
.281	.438	310 LB	220 LB
.500	.625	610 LB	440 LB
.688	1.000	880 LB	
1.125	1.500	1250 LB	
1.750	2.000	1920 LB	

4. CHANGES FROM PREVIOUS ISSUE. MARGINAL NOTATIONS ARE NOT USED IN THIS REVISION TO IDENTIFY CHANGES WITH RESPECT TO THE PREVIOUS ISSUE DUE TO THE EXTENT OF THE CHANGES.

MILITARY INTEREST

Custodians:
Army - AR
Navy - OS
Air Force - 99

Preparing activity:
DLA - IS

(Project 5325-2012-006)

Review activities:
Army – MI
Navy - AS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:59Z
Check the source to verify that this is the current version before use.